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2041: The Next 20 Years

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Technology is the wave of the future and considering all the technological advances that are taking place at such a rapid pace, it's safe to say that the wave of the future is here. What mind-blowing technologies can we expect to see by 2041 that will change the way we think about technology? If we look at the computing industry in the last decade, a lot has changed and some at a very rapid pace. It's exciting to think about what upcoming computers and applications we'll be introduced to and how they'll improve our lives even more in the coming years. We are already witnessing the next phase of technological development, and that phase is convergence.

The current pandemic necessitated the "Remote Work and Learn from Home" mandate. PCs became a must-have device for families with kids at home who no longer could attend classes in person and had to do their studies online or via video conferencing. It is hard to imagine any of us being able to traverse and survive this pandemic without technological means to get through this period of our history. While mobiles and tablets have been useful devices, PCs and laptops have been the workhorses that have become the silent heroes of the Covid-19 pandemic.

What will computing look like in 2041?

As you know, computer usage has changed much more than ever, but the next 20 years already promise to deliver even greater innovation. In modern computers, features such as touchscreens, face recognition, "all-day" battery life, and always-on connectivity are becoming ever more conventional. Today computers are in virtually everything we touch, all day long. We still have an image of computers either on a desk, or these days in our pockets, but computers are in our cars, and refrigerators. Increasingly computers have spread virtually every other device.

The computers of the future will, of course, tap into faster connectivity thanks to 5G and whatever comes after it (well, 6G, 7G, and so on). The next 20 years of computing industry, we will see the blending of Virtual Reality and Artificial Intelligence which will be used everywhere. Joining together VR and AI can provide us with incredible opportunities. It's motivating to think about what type of opportunities the integration of the two will bring us. Which will become so pervasive that nobody will even mention it anymore. It will simply be accepted that every product includes an AI application. One of the biggest differences is that I think that most PCs in 2041 will be smaller and thinner PCs. The way we work has become a lot more adaptable, with or without the pandemic, and we are much more 'on the go'. While this has given rise to the trend 'du jour', which are foldable displays that can offer all the functionalities of a laptop, while giving developers and manufacturers to have freedom to develop experiences which are dynamic and personal.

We have also made a lot of innovations in head-mounted displays (HMD) over the last five years and our future also looks very promising with virtual experiences becoming more mainstream. Using HMD also breaks the limitation of screen size in a much portable form-factor.

With better and faster connectivity, we are already seeing the adoption of cloud storage so we are able to access files seamlessly across devices. Going forward it won't be just files, even processing can move to cloud where performance will be on subscription model as per need. This change will give a lot of flexibility in design as there is less dependency on internal components, heat management etc. as the workload is moved to cloud. This also means cyber security and privacy should be at the forefront of concerns to address by companies and governments as we move forward.

Areas like quantum computing is going to be extremely important in the next two decades; that computers are going to become tiny. This will make a huge difference; nano-computing that you might gulp inside a pill and then will learn about your illness and set about treating it; that brings together genetic computing as well, where we can print parts of the body. If you see, much of the technology that we use today fitted in science fiction movies until recently. Star Trek communicators became mobile phones and Star Trek replicators became 3D printers. The possibilities are limitless, we have come to a point where what we imagine we are able to produce but the important thing is making choices on where we need to innovate and spend energy time and money which can help drive society, economy and humanity forward positively with sustainability at the center.

*Columns may have been edited to fit. Read the original columns on **digit.in***